NOT MEASUREMENT SENSITIVE

MIL-M-63048B(TM)
21 November 1989
SUPERSEDING
MIL-M-63048A(TM)
15 November 1972

MILITARY SPECIFICATION

MANUALS, TECHNICAL: PREPARATION OF EQUIPMENT PUBLICATIONS ON MICROFICHE

This specification is approved for use by the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the layout requirements for the publishing of technical manuals and equipment publications on microfiche.

1.2 Examples/figures. Figures/examples used in this document are for examples only. The text of this document takes precedence over examples.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAMC Materiel Readiness Support Activity, ATTN: AMXMD-MP, Lexington, KY 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC A4850 **AREA TMSS**

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-M-38748	Microfiche: For Engineering/Technical Data, Reports, Studies and Related Data, Requirements for
MIL-M-85337	Manuals, Technical: Quality Assurance Program, Requirements For

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.)

2.1.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents that are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 (DoD adopted) - Packaging, Commercial

(DOD activities can obtain copies of ASTM D 3951 from the Naval Publications and Forms Center, ATTN: NPODS, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Other Government activities, contractors and other private concerns must procure copies from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

Microcopy Resolution Test Chart - NBS 1010A

Copies of NBS 1010A may be obtained from the National Institute of Standards and Technology, Gaithersburg, MD 20899.)

2.2 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Format.** Each microfiche shall consist of a horizontal area along the 148.75mm side, above a grid of image frames arranged in horizontal rows and vertical columns (see figure 1). The grid rows shall be lettered vertically top to bottom with the letters A through G. As specified by the contracting activity, the columns shall be numbered horizontally by placing numerals 1 through 14 in title area or at the bottom as shown on figure 1. The frame identification shall be placed on the bottom center of each image frame.

3.2 **Layout.** A layout shall be prepared in accordance with figure 1 and the following instructions. Figure 1 indicates order of precedence for title area, resolution chart, and front matter. The order of precedence for the remaining frames shall be indicated on the layout sheet.

3.2.1 **Title area.** The title area for microfiche shall include the following. Each marking shall be readable by the unaided eye.

a. The basic publication number on the left-hand corner, first line of title area, with the publication date placed beneath the number.

b. The title of the basic publication at the center of the title area. Abbreviated titles may be used to conserve space.

c. Microfiche number on the top line, right-hand corner of the title area.

d. A statement, placed beneath t+he microfiche number, indicating in which grid area the table of contents is located.

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e. If applicable, appropriate change incorporation statement after the date of the publication (see 3.2.1.3).

f. If applicable, a revision symbol (see 3.2.1.3) before the publication number and the date of revision immediately below the publication number.

g. If specified by the contracting activity, numerals 1 through 14 at bottom of title area to identify columns (see 3.1).

h. On the first microfiche prepared for a publication, the header information for classification, distribution limitation statement, export warning, and destruction notice, as applicable, at the bottom of the title area.

3.2.1.1 **Microfiche number.** If the entire publication can be contained on one microfiche, the microfiche shall be numbered "FICHE NO. 1 OF 1". If additional microfiche are needed, they shall be numbered consecutively in order of preparation (such as FICHE NO. 1 OF 4, FICHE NO. 2 OF 4, etc). See figure 1.

3.2.1.2 **Changes and revisions.** No individual page changes shall be prepared on microfiche. All changes, including those published as page changes in the hard copy version of the manual, shall be incorporated and published as a revision(s) to the affected microfiche.

3.2.1.3 **Change/revision identification.** Each microfiche sheet which is a revision of a previously published microfiche edition shall be identified by placing an asterisk in front of the publication's number. The date of the revision or change shall be placed immediately below the publication number. If applicable, the date shall be followed by an appropriate change incorporation statement as shown in the following example.

*TM 9-1415-251-34
16 Jan 89, incorporates Changes
1 and 2

3.2.1.4 **Security classification header.** Each set of classified microfiche shall bear the same classification as the original publication. Each microfiche shall be marked with the classification of the highest classified frame of its set. The appropriate security classification of the set shall be indicated by boldface capital letters located in the title area of each microfiche. The security marking of all camera-ready copy from which microfiche is to be made shall be as specified in MIL-M-38784.

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3.2.1.5 **Distribution limitation statement.** The first microfiche prepared for a publication shall be marked with the appropriate abbreviated distribution limitation statement. This marking shall follow the classification marking in the title area, as shown in the following example.

Sample header - Revised DMWR

*USAMICOM DMWR 9-1400-250-8 JANUARY 1973	MOTORS AND ROTARY DEVICES	FICHE NO. 2 OF 3 FOR TABLE OF CONTENTS SEE FRAME A-4
UNCLASSIFIED DISTRIBUTION C:	EXPORT WARNING	DESTRUCTION NOTICE

3.2.1.6 **Export warning.** If applicable, the first microfiche shall contain the words "EXPORT WARNING" immediately following the abbreviated distribution limitation statement.

3.2.1.7 **Destruction notice.** The first microfiche shall contain the words "DESTRUCTION NOTICE" immediately following the export warning or the abbreviated distribution limitation statement, as applicable.

3.2.2 **Resolution chart and reduction ratio.** The NBS 1010A Microcopy Resolution Chart and microfiche reduction ratio (eg, 24:1) shall be located in grid area A-1 of each microfiche. The resolution chart shall be so placed that it can be read from left to right. The vertical lines of the chart shall be parallel to those of the microfiche grid.

3.2.3 **Distribution limitation, export warning, and destruction notice.** The distribution limitation, export warning, and destruction notice statements shall be placed in grid area A-2 on the first microfiche. The statements shall be in accordance with MIL-M-38784.

3.2.4 **Security downgrading/security marking information.** Security marking information (see figure 2) or appropriate security downgrading statement shall appear in grid area A-2 on the first microfiche, following the distribution limitation, export warning, and destruction notice statements (see 3.2.2)

3.2.5 **Warning page.** The warning page, in accordance with MIL-M-38784, shall appear in grid area A-3 on the first microfiche.

3.2.6 **Title block page.** The title block page shall appear in grid area A-4 of the first microfiche.

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3.2.6.1 **Reporting errors and recommending improvements statement.** The applicable reporting errors and recommending improvements statement (boxed -in) shall appear below the title block. The address of the responsible proponent shall be inserted in this box.

a. **Unclassified TMs.** For unclassified TMs, the following statement shall be used:

"REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms), direct to: (address of proponent). A reply will be furnished to you."

b. **Classified TMs.** For classified TMs, the following statement shall be used:

"REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes or if you know of a way to improve this manual, write and tell us about it. Address your correspondence to (insert address of the proponent). When dealing with classified information, make sure that your correspondence is properly marked and is handled in accordance with AR 380-5."

3.2.6.1.1 **Table of contents.** When space permits, the table of contents shall begin below the reporting errors and recommending improvements statement. If space does not permit, it shall begin in the next frame. The table of contents shall be in accordance with MIL-M-38784 except that an additional column showing the microfiche and frame numbers for each entry shall be added.

3.2.7 **Illustrations.** All illustrations shall be prepared or converted to a vertical format presentation.

3.2.8 **Foldouts.** Foldouts shall be planned to cover as many adjacent frames (horizontal and vertical) as may be required by the reduction ratio and to facilitate use in the reader. They may be repeated on separate microfiche having relevant text. Pages shall be so filmed that, when mounted, they are in a position to be read from left to right without being rotated in the reader. Vertical foldups shall be used only if no other method can adequately present this information.

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3.2.9 **References.** References to publications other than the one being microfiched shall be in accordance with the requirements of MIL-M-38784. Material from publications other than the one being microfiched which will occupy one frame or less shall be included rather than referenced. References to other portions of the same microfiched document shall be made by either paragraph number, sheet and frame number, or a combination thereof.

3.2.10 **Image sequence.** The image frames of the publication pages shall be arranged to follow the order of presentation indicated by figure 1. The detail arrangement shall follow the worksheet layout plan. One or more blank images may be left at the end of each horizontal line of each sheet, or one or more blank images may be left after each major segment (eg, front matter, each chapter, section, or appendix, etc).

3.3 **Container/envelope markings.** As specified by the contracting activity, an appropriate distribution limitation statement, export warning, classification, and destruction notice shall be placed on the front on the microfiche container/envelope. Placement of the markings shall be as shown on figure 3.

3.4 **Production of microfiche.** Microfiche shall be produced as specified in MIL-M-38748.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance

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comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Quality assurance provision.** Quality assurance in the preparation of microfiche shall be the responsibility of the contractor and shall be implemented in accordance with MIL-M-85337 to the extent specified by the contracting activity.

5. PACKAGING

5.1 **Packaging requirements.** Unless otherwise specified by the contracting activity, the packaging and packing of the microfiche products shall conform to ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** This document prescribes the production of equipment manuals on microfiche. The technical content requirements for the manual shall be prescribed by appropriate technical manual content specification(s).

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Location of column numbering (see 3.1).

6.3 **Subject term (key word) listing.**

Image frame
Resolution

6.4 **Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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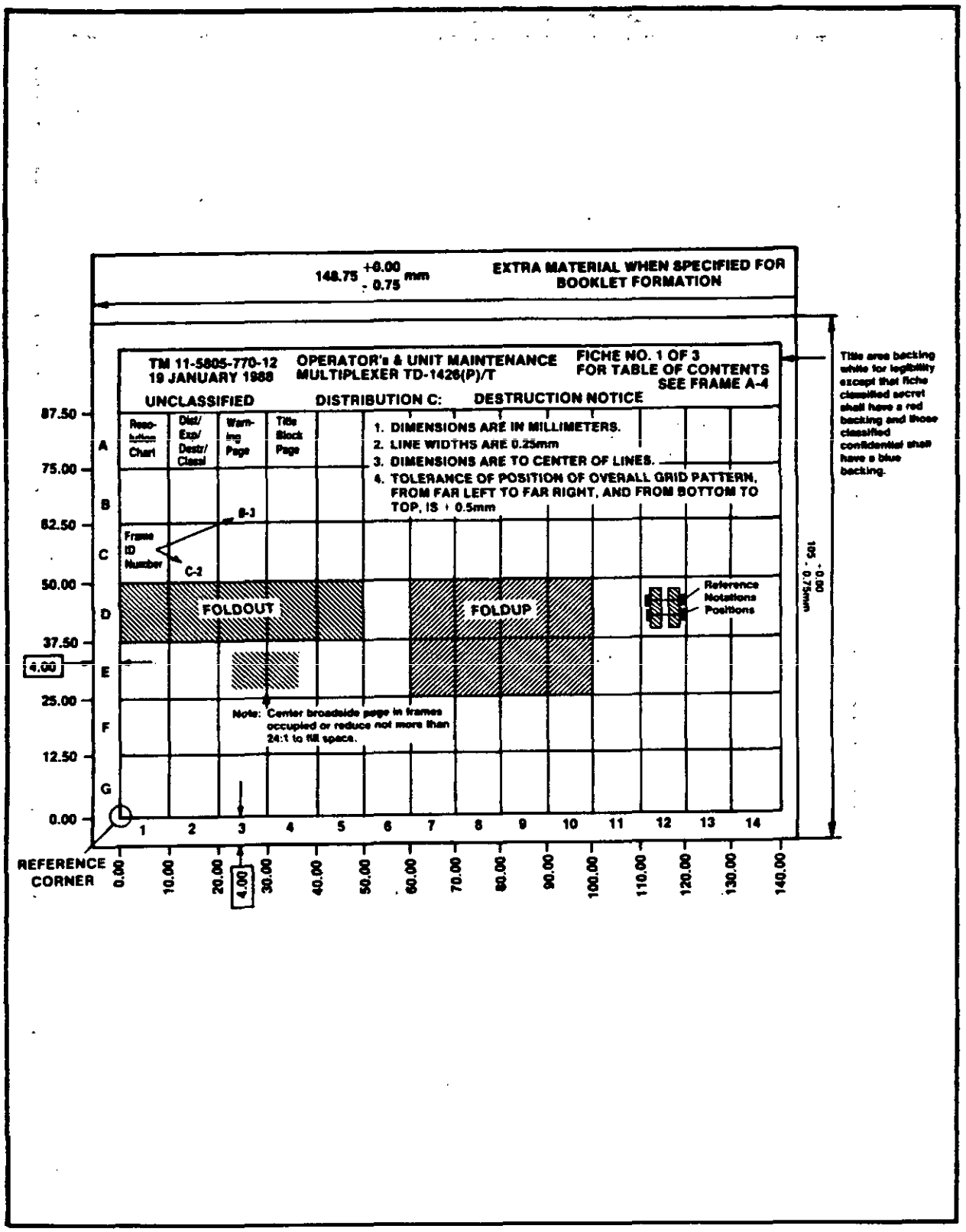


FIGURE 1. Example of microfiche format and layout.

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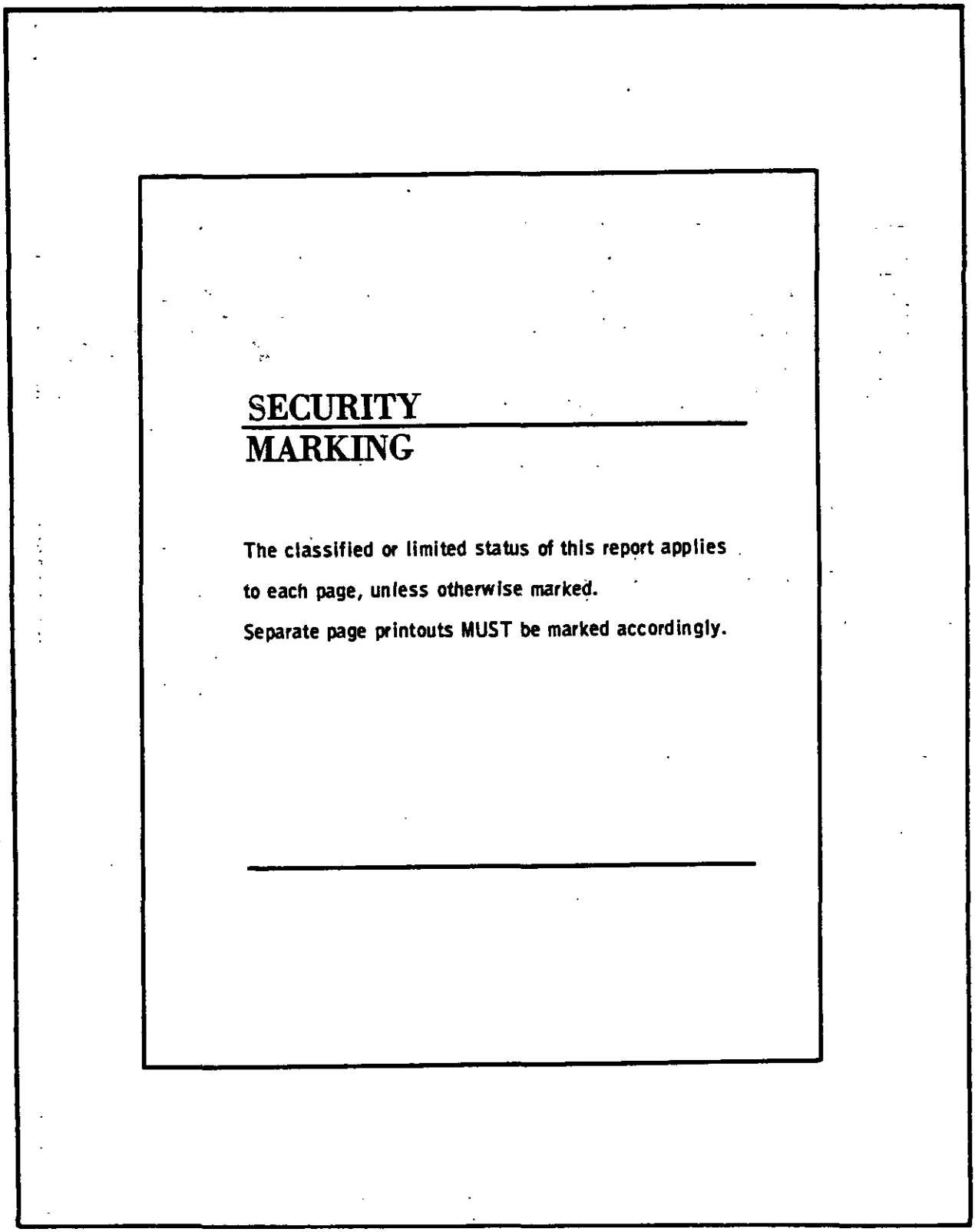


FIGURE 2. Example of security marking information.

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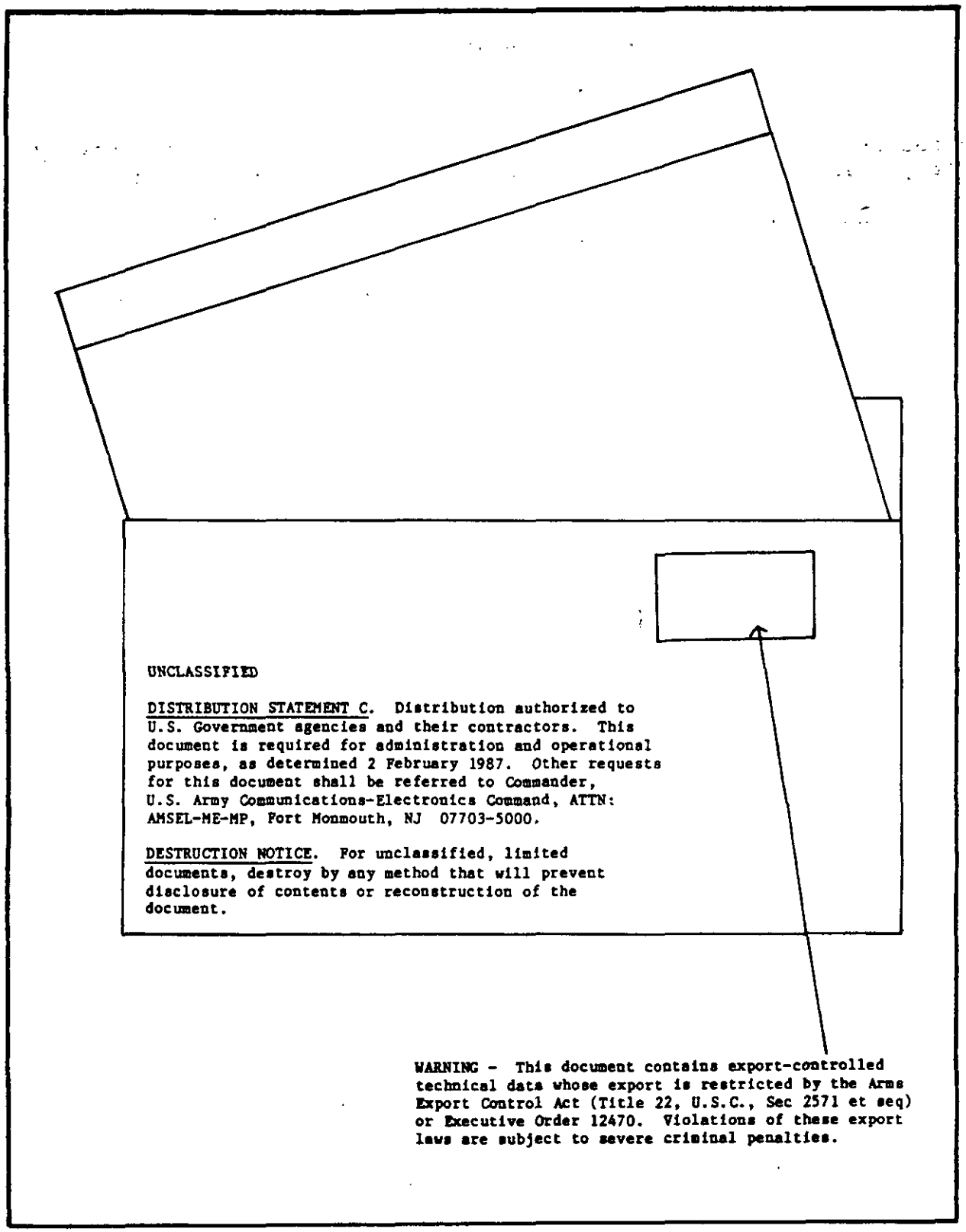


FIGURE 3. Example of container/envelope markings.

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Custodian:
Army - TM

Review activities:
Army - AT, AV, CR, ER, ME, MI, SC

Preparing activity:
Army - TM

(Project TMSS-A249)

*U.S. GOVERNMENT PRINTING OFFICE: 1990--704-034/30252

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
Mil-M-63048B(TM)	Manuals, Technical: Preparation of Equipment Publications on Microfiche

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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